

Unit 114 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 116.3 ac, BAF 20 variable-radius cruise on a 300 ft grid, 56 plots, 593 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	212 sq ft/ac (SE 2.1 %)	Type	Sierran mixed conifer
Trees per acre	93	CWHR size / density	4 / D (cover 85 %)
QMD	20.5 in	SDI	276 (48 % of max 573)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	208 sq ft/ac 21.6 in	BA QMD	4 sq ft/ac 7.9 in
TPA	81	TPA	11
Volume	43.4 MBF/ac (7899 cu ft)	Volume	2.1 green tons/ac (69 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~154 sq ft/ac BA	Sampling error	4.3 % at 95 % (t = 2.00, n = 56)
Species (BA)	PP 65, WF 61, IC 33, DF 30, SP 22	Stratum standard	40 % - MEETS; 56 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	5	7.0	1.8	28	0.2
8-11.9	23	13.4	8.2	173	1.0
12-15.9	70	22.5	25.0	672	3.7
16-19.9	78	15.8	27.9	854	4.7
20-23.9	103	14.3	36.8	1254	6.9
24-27.9	70	7.0	25.0	930	5.1
28-31.9	79	5.7	28.2	1189	6.5
32-35.9	59	3.4	21.1	951	5.2
36-39.9	30	1.3	10.7	501	2.8
40-43.9	25	0.9	8.9	433	2.4
44-47.9	21	0.7	7.5	389	2.1
48-51.9	9	0.2	3.2	168	0.9
52-55.9	5	0.1	1.8	98	0.5
56-59.9	7	0.1	2.5	128	0.7
60-63.9	9	0.2	3.2	200	1.1
All	593	92.8	211.8	7968	43.8

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	183	32.0	65.4	19.4	2398	13.2
WF	172	25.3	61.4	21.1	2370	13.0
IC	92	13.8	32.9	20.9	1233	6.8
DF	84	12.6	30.0	20.9	1153	6.3
SP	62	9.1	22.1	21.1	815	4.5
All	593	92.8	211.8	20.5	7968	43.8

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 114: plots on the 300 ft grid (orange = QA flag)

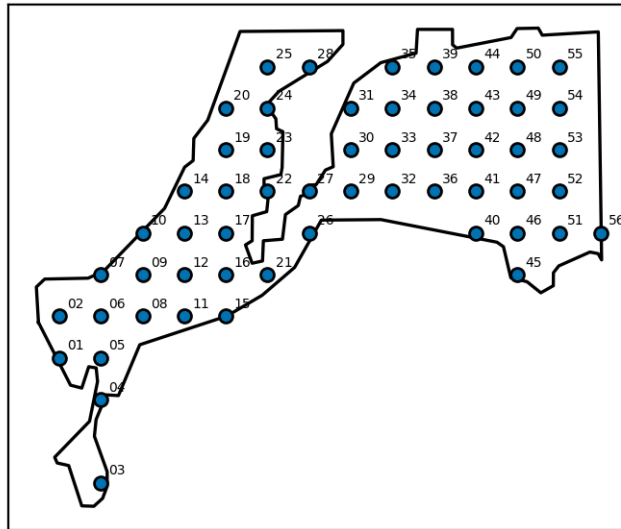


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
114-01	12	240	120	ok
114-02	9	180	15	ok
114-03	8	160	82	ok
114-04	12	240	117	ok
114-05	9	180	39	ok
114-06	11	220	60	ok
114-07	10	200	87	ok
114-08	14	280	109	ok
114-09	11	220	60	ok
114-10	12	240	100	ok
114-11	12	240	62	ok
114-12	12	240	126	ok
114-13	9	180	41	ok
114-14	11	220	33	ok
114-15	10	200	122	ok
114-16	11	220	63	ok
114-17	11	220	45	ok
114-18	10	200	15	ok
114-19	13	260	23	ok
114-20	12	240	76	ok
114-21	12	240	24	ok
114-22	9	180	54	ok
114-23	12	240	36	ok
114-24	8	160	16	ok
114-25	12	240	132	ok
114-26	9	180	27	ok
114-27	9	180	21	ok
114-28	11	220	420	ok
114-29	8	160	26	ok
114-30	12	240	28	ok
114-31	10	200	56	ok
114-32	10	200	75	ok
114-33	12	240	37	ok
114-34	11	220	55	ok
114-35	8	160	52	ok
114-36	6	120	28	ok
114-37	13	260	42	ok
114-38	12	240	155	ok
114-39	11	220	22	ok
114-40	11	220	75	ok
114-41	10	200	62	ok
114-42	11	220	39	ok
114-43	10	200	41	ok
114-44	10	200	42	ok
114-45	14	280	143	ok
114-46	10	200	75	ok
114-47	12	240	56	ok
114-48	8	160	39	ok
114-49	7	140	34	ok
114-50	10	200	31	ok
114-51	10	200	18	ok

Plot	Trees in	BA sq ft/ac	Slope %	Status
114-52	10	200	24	ok
114-53	12	240	38	ok
114-54	10	200	26	ok
114-55	11	220	23	ok
114-56	13	260	19	ok